

MARBOXIN 10% w/v INJECTION

Description and Composition

MARBOXIN 10% w/v INJECTION is a yellow or straw coloured solution. Each ml contains 100mg of Marbofloxacin as its active ingredient and 10mg Benzyl Alcohol as its preservative.

Pharmacodynamics

Marbofloxacin is a synthetic, bactericidal antimicrobial, belonging to the fluoroquinolone group which acts by inhibition of DNA gyrase. It is effective against a wide range of Gram positive bacteria (in particular *Staphylococci*) and Gram negative bacteria (*Escherichia coli*, *Salmonella typhimurium*, *Citrobacter freundii*, *Enterobacter cloacae*, *Serratia marcescens*, *Morganella morganii*, *Proteus* spp, *Klebsiella* spp, *Shigella* spp, *Pasteurella* spp, *Haemophilus* spp, *Moraxella* spp, *Pseudomonas* spp) as well as *Mycoplasma* spp.

Pharmacokinetics

After subcutaneous or intramuscular administration in cattle and intramuscular administration in pigs at the recommended dose of 2mg/kg, marbofloxacin is readily absorbed and reaches maximal plasma concentrations of 1.5µg/ml within less than 1 hour. Its bioavailability is close to 100%.

It is weakly bound to plasma proteins (less than 10% in pigs and 30% in cattle), extensively distributed and in most tissues (liver, kidney, skin, lung, bladder, uterus, digestive tract) it achieves a higher concentration than in plasma.

In cattle, marbofloxacin is eliminated slowly in pre-ruminating calves ($t_{1/2\beta} = 5-9h$) but faster in ruminant cattle ($t_{1/2\beta} = 4-7h$) predominantly in the active form in urine (3/4 in pre-ruminating calves, 1/2 in ruminants) and faeces (1/4 in pre-ruminating calves, 1/2 in ruminants).

In pigs, marbofloxacin is eliminated slowly ($t_{1/2\beta} = 8-10h$) predominantly in the active form in urine (2/3) and faeces (1/3).

Indication

Cattle: Indicated in the treatment of respiratory infections caused by susceptible strains of organisms. It is also indicated in the treatment of acute *E. coli* mastitis.

Pigs: Indicated in the treatment of Metritis Mastitis Agalactia syndrome caused by susceptible strains of organisms.

Recommended Dose

The recommended dosage is 2mg per kg per day (1ml per 50kg) in a single daily injection by intramuscular, subcutaneous or intravenous routes in cattle and by intramuscular route in pigs.

Treatment durations is 3 days in pigs and 3 to 5 days in cattle.

In order to reduce the risk of particulate contamination of the product, it is recommended that a draw-off needle be used to reduce the number of times the septum is punctured.

Route of Administration

To be given by intramuscular, subcutaneous or intravenous injection in cattle.

To be given by intramuscular injection in pigs.

Contraindications

None known.

Warning and Precautions

Special warnings for each target species:

None.

Special precautions for use**Special precautions for use in animals:**

Official and local antimicrobial policies should be taken into account when the product is used. Fluoroquinolones should be reserved for the treatment of clinical conditions which have responded poorly, or are expected to respond poorly, to other classes of antimicrobials. Whenever possible, fluoroquinolones should only be used based on susceptibility testing. Use of the product deviating from the instructions may increase the prevalence of bacteria resistant to the fluoroquinolones and may decrease the effectiveness of treatment with other quinolones due to the potential for cross resistance.

Special precautions to be taken by the person administering the veterinary medicinal product to animals:

People with known hypersensitivity to fluoroquinolones should avoid using this product.

Wash hands after use.

Interaction with Other Medicaments

None known.

Pregnancy and Lactation

May be used in pregnant and lactating cows and sows.

Side Effects

No severe side-effects are to be expected at doses up to 3 or 5 times the recommended dose in cattle and pigs respectively. In particular, no lesions of the articular joints are encountered.

Intramuscular or subcutaneous injections are well tolerated although transitory inflammatory lesions without clinical impact can occur at the injection site.

Symptoms and Treatment of Overdose

Overdosage may cause acute signs in the form of neurological disorders which would have to be treated symptomatically.

Storage Condition

Store below 30°C. Protect from light.

Shelf life

3 years.

Following withdrawal of first dose, use the product within 24 hours. Any unused material should be discarded.

Withdrawal Periods

Meat: Pigs: 4 days, Cattle: 6 days

Milk: 36 hours

Packing

20ml, 50ml, 100ml, 250ml.

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	Manufacturer & Product Registration Holder
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